FEDERAL COURT OF AUSTRALIA

Coal of Queensland Pty Ltd v Innovation and Science Australia [2021] FCAFC 54

Appeal from:	*Coal of Queensland Pty Ltd and Innovation and Science Australia* (Taxation) [2020] AATA 126 (30 January 2020)
File number:	QUD 51 of 2020
Judgment of:	**LOGAN, GRIFFITHS AND MOSHINSKY JJ**
Date of judgment:	23 April 2021
Catchwords:	**TAXATION** – research and development tax offset – where applicant held exploration permit for coal – where coal in the area had high level of ash content and the coal seams were banded – where applicant commenced a series of activities to investigate the nature and economic viability of mining the coal – where the Tribunal found that none of the registered activities were "core R&D activities" within the meaning of s 355-25(1) of the *Income Tax Assessment Act 1997* (Cth) – where the Tribunal found that the exclusion in s 355-25(2)(b) applied – where the Tribunal found that the activities were not "supporting R&D activities" within s 355-30 – whether the Tribunal made findings for which there was 'no evidence' – whether the Tribunal applied the wrong legal test – held: appeal dismissed
Legislation:	*Administrative Appeals Tribunal Act 1975* (Cth), s 44 *Income Tax Assessment Act 1997* (Cth), ss 355-1, 355-5, 355-25, 355-30, 355-35 *Industry Research and Development Act 1986* (Cth), ss 4, 27A, 27F, 27J, 27K, 27L, 30D, 30E *Mineral Resources Act 1989* (Qld)
Cases cited:	*Australian Broadcasting Tribunal v Bond* (1990) 170 CLR 321 *Commissioner of Taxation v Bogiatto* [2020] FCA 1139 *Comptroller-General of Customs v Pharm-A-Care Laboratories Pty Ltd* (2020) 375 ALR 98 *Docklands Science Park Pty Ltd and Innovation Australia* [2015] AATA 973 *Haritos v Federal Commissioner of Taxation* (2015) 233 FCR 315

	Keith v Gal [2013] NSWCA 339 *Kostas v HIA Insurance Services Pty Ltd* (2010) 241 CLR 390 *Li v Attorney General (NSW)* (2019) 99 NSWLR 630 *Minister for Immigration and Multicultural Affairs v Al-Miahi* (2001) 65 ALD 141 *Monie v Commonwealth* (2005) 63 NSWLR 726 *Moreton Resources Ltd v Innovation and Science Australia* (2019) 271 FCR 211 *Re DBTL and Innovation Australia* (2013) 137 ALD 88 *Rix's Creek Pty Ltd; Bloomfield Collieries Pty Ltd and Innovation Australia* [2017] AATA 645
Division:	General Division
Registry:	Queensland
National Practice Area:	Taxation
Number of paragraphs:	155
Date of hearing:	23 November 2020
Counsel for the Applicant:	Mr AT Broadfoot QC with Mr AJ Bailey
Solicitor for the Applicant:	Holding Redlich
Counsel for the Respondent:	Mr SJ Free SC with Ms C Trahanas
Solicitor for the Respondent:	Clayton Utz

ORDERS

QUD 51 of 2020

BETWEEN:	**COAL OF QUEENSLAND PTY LTD (ACN 149 250 939)** Applicant
AND:	**INNOVATION AND SCIENCE AUSTRALIA** Respondent

ORDER MADE BY: **LOGAN, GRIFFITHS AND MOSHINSKY JJ**

DATE OF ORDER: **23 APRIL 2021**

THE COURT ORDERS THAT:

1. The applicant have leave to amend the amended notice of appeal to the form of the proposed further amended notice of appeal (second version) provided to the Court by email at approximately 3.44 pm on 23 November 2020.
2. The applicant forthwith file and serve the further amended notice of appeal referred to in paragraph 1.
3. The appeal be dismissed.
4. The applicant pay the respondent's costs of the appeal, as agreed or taxed.

Note: Entry of orders is dealt with in Rule 39.32 of the *Federal Court Rules 2011*.

REASONS FOR JUDGMENT

THE COURT:

Introduction

1 The applicant holds an exploration permit for coal covering approximately 120 square kilometres in the Bowen basin area in Queensland. The activities carried on by the applicant in the area covered by the exploration permit are known as the Wilton Project. The coal deposits found in the Wilton Project include part of what is known as the Fort Cooper Coal Measures (the **FCCM**). The coal found in the FCCM has a relatively high level of ash content, and the coal seams are banded. These characteristics mean the Wilton Project has not been commercially viable because of the cost and difficulty of using established methods for liberating the coal.

2 In about July 2011, the applicant formed the view that it might be possible to produce a viable coking coal product from the FCCM notwithstanding those characteristics. To that end, it commenced a series of activities to investigate the nature and economic viability of mining the FCCM deposit.

3 On 30 April 2013, the applicant applied to register certain activities conducted during the year ended 30 June 2012 (the **2012 year**) as research and development (**R&D**) activities under applicable legislation. The relevant legislative provisions are located in Div 355 of the *Income Tax Assessment Act 1997* (Cth) (the **ITAA 1997**) and in the *Industry Research and Development Act 1986* (Cth) (the **IR&D Act**). The activities were said to form part of a project referred to as "Design and development of a new mining and beneficiation process" (the **Project**). The applicant applied to register four activities as "core R&D activities" and one activity as a "supporting R&D activity". The definitions of these expressions are set out later in these reasons.

4 The activities were registered by the respondent, which is referred to as the "Board" in the IR&D Act (the **Board**).

5 Subsequently, the Board conducted a compliance review. On 28 May 2015, a delegate of the Board decided that the applicant's registered activities for the 2012 year were neither core R&D activities nor supporting R&D activities. Following a process of internal review, that decision was confirmed by the Board on 27 January 2017 (the **internal review decision**).

6 The applicant applied to the Administrative Appeals Tribunal (the **Tribunal**) for review of the internal review decision. The hearing took place over three days in August 2018. On 30 January 2020, the Tribunal decided to affirm the internal review decision and provided reasons for decision (the **Reasons**). The Tribunal found that the activities registered as core R&D activities were not "core R&D activities", as defined, because they were not carried out by applying a systematic progression of work based on principles of established science, and because they did not proceed from hypothesis to experiment, observation and evaluation, leading to logical conclusions. The Tribunal also found that the activities were not conducted for the purpose of generating new knowledge, being an element of the definition of "core R&D activities". Further, the Tribunal found that the activities came within the exception in s 355-25(2)(b) of the ITAA 1997 (set out later in these reasons). The Tribunal rejected an alternative contention put forward by the applicant that certain of the activities were "supporting R&D activities".

7 Pursuant to s 44 of the *Administrative Appeals Tribunal Act 1975* (Cth) (the **AAT Act**), the applicant 'appeals' to this Court, on a question of law, from the decision of the Tribunal. The Court's original jurisdiction in this matter is being exercised by a Full Court.

8 The applicant's current notice of appeal is its amended notice of appeal from a tribunal dated 26 October 2020. At the hearing of the appeal, the applicant sought leave to amend the amended notice of appeal. A proposed further amended notice of appeal was provided to the Court by email during the hearing (which took place by video using Microsoft Teams). Following some discussion, the applicant prepared and provided to the Court by email a second version of the proposed further amended notice of appeal. We heard argument on the proposed additional question of law and the associated additional grounds, and reserved our decision on whether to grant leave to amend.

9 In broad terms, by its existing and proposed additional grounds, the applicant challenges many of the factual findings made by the Tribunal, either on the basis that the finding was "not open" or because the Tribunal applied the wrong legal test. Insofar as the applicant contends that certain factual findings were "not open", there is an issue whether the applicant's grounds raise a question of law, as required by s 44 of the AAT Act. The applicant also contends that the Tribunal failed to give adequate reasons and failed to make findings on material questions of fact.

10 For the reasons that follow, in our view, the applicant should be granted leave to amend the amended notice of appeal, but the appeal should be dismissed.

Key legislative provisions

11 The key legislative provisions were recently set out in the judgment of the Full Court in *Moreton Resources Ltd v Innovation and Science Australia* (2019) 271 FCR 211 (***Moreton***), which also concerned the 2012 year (as well as other years). In the following paragraphs, we draw on the Full Court's summary of the provisions.

The ITAA 1997

12 The relevant provisions of the ITAA 1997 are contained in Div 355 (Research and Development). Section 355-1 explains that an R&D entity may be entitled to a tax offset for R&D activities; to be entitled to the tax offset, the R&D entity needs one or more notional deductions under the Division. Section 355-1 also explains that there are two kinds of notional deductions: one is for expenditure on R&D activities; the other is for the decline in value of tangible depreciating assets used for R&D activities. A note under s 355-1 states that all of these notional deductions require the R&D entity to be registered for the R&D activities under Pt III of the IR&D Act.

13 The expression "R&D entity" is defined in s 355-35. The definition includes a body corporate incorporated under Australian law.

14 The object of Div 355 is set out in s 355-5:

> **355-5 Object**
>
> (1) The object of this Division is to encourage industry to conduct research and development activities that might otherwise not be conducted because of an uncertain return from the activities, in cases where the knowledge gained is likely to benefit the wider Australian economy.
>
> (2) This object is to be achieved by providing a tax incentive for industry to conduct, in a scientific way, experimental activities for the purpose of generating new knowledge or information in either a general or applied form (including new knowledge in the form of new or improved materials, products, devices, processes or services).

15 The expression "R&D activities" is defined in s 355-20 to mean "core R&D activities" or "supporting R&D activities".

16 "Core R&D activities" is defined in s 355-25:

> **355-25** ***Core R&D activities***
>
> (1) ***Core R&D activities*** are experimental activities:
> (a) whose outcome cannot be known or determined in advance on the basis of current knowledge, information or experience, but can only be determined by applying a systematic progression of work that:
> (i) is based on principles of established science; and
> (ii) proceeds from hypothesis to experiment, observation and evaluation, and leads to logical conclusions; and
> (b) that are conducted for the purpose of generating new knowledge (including new knowledge in the form of new or improved materials, products, devices, processes or services).
>
> (2) However, none of the following activities are ***core R&D activities***:
> (a) market research, market testing or market development, or sales promotion (including consumer surveys);
> (b) **prospecting, exploring or drilling for minerals or *petroleum for the purposes of one or more of the following:**
> **(i) discovering deposits;**
> **(ii) determining more precisely the location of deposits;**
> **(iii) determining the size or quality of deposits;**
> (c) management studies or efficiency surveys;
> (d) research in social sciences, arts or humanities;
> (e) commercial, legal and administrative aspects of patenting, licensing or other activities;
> (f) activities associated with complying with statutory requirements or standards, including one or more of the following:
> (i) maintaining national standards;
> (ii) calibrating secondary standards;
> (iii) routine testing and analysis of materials, components, products, processes, soils, atmospheres and other things;
> (g) any activity related to the reproduction of a commercial product or process:
> (i) by a physical examination of an existing system; or
> (ii) from plans, blueprints, detailed specifications or publically available information;
> (h) developing, modifying or customising computer software for the dominant purpose of use by any of the following entities for their internal administration (including the internal administration of their business functions):
> (i) the entity (the ***developer***) for which the software is developed, modified or customised;
> (ii) an entity *connected with the developer;
> (iii) an *affiliate of the developer, or an entity of which the developer is an affiliate.
>
> (Emphasis added.)

17 "Supporting R&D activities" is defined in s 355-30:

> **355-30** ***Supporting R&D activities***
>
> (1) ***Supporting R&D activities*** are activities directly related to *core R&D

activities.

(2) However, if an activity:
(a) is an activity referred to in subsection 355-25(2); or
(b) produces goods or services; or
(c) is directly related to producing goods or services;
the activity is a ***supporting R&D activity*** only if it is undertaken for the dominant purpose of supporting *core R&D activities.

18 Subdivision 355-D deals with notional deductions for R&D expenditure. In broad terms, the effect of this Subdivision is that an R&D entity can notionally deduct its expenditure on registered R&D activities for which certain conditions are met.

The IR&D Act

19 Section 4 of the IR&D Act provides that the terms "core R&D activities" and "supporting R&D activities" have the same meaning as in the ITAA 1997.

20 Registration of an R&D entity for R&D activities is dealt with in Pt III of the IR&D Act. Under that Part (in particular, Div 2 of that Part):

(a) the Board may register an R&D entity for R&D activities conducted during an income year;

(b) the Board may make findings about the nature of an R&D entity's activities, both before and after registration; and

(c) such findings bind the Commissioner for the purposes of any entitlement of the R&D entity to a tax offset under Div 355 of the ITAA 1997 for the activities.

21 Section 27A of the IR&D Act deals with registering R&D entities for R&D activities. Section 27F(1) provides that the Board may conduct one or more examinations of all or part of an R&D entity's registration under s 27A for an income year for the purposes of making one or more findings under s 27J(1) (set out below). The note under s 27F(1) explains that a finding under s 27J(1) will support the entity's registration, or cause the variation of that registration.

22 Section 27J provides as follows:

27J Findings about a registration

(1) The Board may make one or more findings to the following effect about an R&D entity's registration under section 27A for an income year (the ***registration year***):
(a) that all or part of a registered activity was a core R&D activity conducted during the registration year;

(b) that all or part of a registered activity was not an activity of a kind covered by paragraph (a);

(c) that all or part of a registered activity was a supporting R&D activity conducted during the registration year and in relation to:

(i) one or more specified registered core R&D activities; or

(ii) one or more specified core R&D activities for which the entity has been registered in an earlier income year; or

(iii) one or more specified core R&D activities yet to be conducted for which the entity could be registered in the registration year if those activities were conducted during the registration year; or

(iv) several specified core R&D activities, each covered by subparagraph (i), (ii) or (iii);

(d) that all or part of a registered activity was not an activity of a kind covered by paragraph (c).

Note 1: A finding is reviewable (see Division 5).

Note 2: The Board could make a finding under paragraph (b) if, for example, the Board has insufficient information to make a finding under paragraph (a). Similarly, the Board could make a finding under paragraph (d) if it has insufficient information to make a finding under paragraph (c).

(2) If the Board makes a finding under subsection (1) in relation to the R&D entity's registration, the Board may specify in the finding the times to which the finding relates.

Example: A finding under paragraph (1)(a) could specify the times during the registration year that a registered activity was a core R&D activity.

(3) This section has effect subject to section 32B (findings cannot be inconsistent with any earlier findings).

The effect of this provision is that the Board may make findings as described in paragraph (a), (b), (c) or (d) of subsection (1) in respect of registered activities. For example, if an activity is registered for a particular income year, the Board may make a finding pursuant to paragraph (a) that the activity was a "core R&D activity" and that it was conducted during the income year. By way of further example, if an activity is registered for a particular income year, the Board may make a finding pursuant to sub-paragraph (c)(ii) that the activity was a "supporting R&D activity" conducted in the income year, in relation to one or more specified "core R&D activities" for which the entity was registered in an earlier income year. In principle, if activities are registered as "core R&D activities", the Board can nevertheless find that they are "supporting R&D activities*"* pursuant to paragraph (c).

23 Section 27K deals with notice of the Board's findings, or of decisions refusing to make findings. Section 27L deals with automatic variations so that an R&D entity's registration is consistent with the Board's findings.

24 Div 5 of Pt III of the IR&D Act deals with review of decisions of the Board. Section 30D provides for a process of internal review. Section 30E deals with external review by the Tribunal of internal review decisions.

Background facts and matters

25 The following statement of background facts and matters (including evidence before the Tribunal) is substantially based on the Reasons. In this section, we have not drawn on factual findings that are the subject of challenge by the applicant in its amended notice of appeal or proposed further amended notice of appeal (second version).

General matters

26 Since 2011, the applicant has held an exploration permit for coal issued pursuant to the *Mineral Resources Act 1989* (Qld) (EPC1235). As noted above, the permit covers approximately 120 square kilometres in the Bowen basin area in Queensland. The permit area is approximately 40 km north east of the town of Emerald. The activities carried on by the applicant in the area of EPC1235 are known as the Wilton Project.

27 As noted above, the coal deposits found in the Wilton Project include part of the FCCM. The FCCM is a body of coal that sits between the Rangal Coal Measures and the Moranbah Coal Measures, which are highly profitable due to the high quality of the coal and the ease and relative low cost at which the coal can be mined and processed. The coal found in the FCCM, in contrast, has a relatively high level of ash content, and the coal seams are banded. The characteristics of the FCCM mean the Wilton Project has not been commercially viable because of the cost and difficulty of using established methods for liberating the coal.

28 In about July 2011, the applicant formed the view that it might be possible to produce a viable coking coal product from the FCCM notwithstanding the characteristics referred to above. To that end, the applicant commenced a series of activities to investigate the nature and economic viability of mining the FCCM deposit. The applicant's vision of new ways to mine and process coal would only be realised, if at all, over an extended period.

29 The Tribunal noted at [37] of the Reasons that the applicant relied on evidence from five lay witnesses and one expert witness, Dr Jeffrey Euston. The respondent relied on the evidence of one expert witness, Dr Andrew Vince.

Mr Vorias's evidence

30 Mr Cameron Vorias has been the director of the applicant since July 2011. As a director, he was actively involved in monitoring the progress of the relevant work. Upon joining the applicant in July 2011, and following the provision of a further qualified geologist's statement prepared by Dr LeBlanc Smith in March 2011, Mr Vorias saw a real opportunity to extract good quality coking coal from the FCCM that had previously been considered uneconomic. However, this could only be achieved if the applicant made improvements to existing mining and beneficiation processes.

31 In his witness statement dated 20 October 2017, Mr Vorias stated that in order to produce coal that was suitable for the international market, it would be necessary to get the coal extracted from the Wilton Project to a specific ash level of less than 12%. He stated the main challenge was to get the coal to this acceptable level of ash impurity, at the right size, and without excessive moisture or other impurities. Mr Vorias outlined the hypothesis at [37] of his statement as follows:

> [B]y the end of around July 2011, I (and I believe my fellow directors) had recognised that it might theoretically be possible for improvements to be made to existing coal mining and beneficiation processes, or new processes could be developed, that would enable the mining, processing and handling (transporting) of highly banded coal seams known to exist in the Bowen basin (such as the FCCM) and elsewhere to be achieved economically by achieving a combination of (a) the extraction of material with a higher yield of coal to waste than had been done previously, (b) the development of a crushing and washing plant configuration that would operate differently from those that had previously been used and that (c) would result in a product able to be handled and transported, which had not previously been thought possible. We recognised that if such improved or new processes could be developed, it might be possible to develop these types of deposits which had historically been regarded as unattractive by major mining companies … because they were thought to be uneconomic due to high levels of ash and waste being present and the consequent need (based on existing technologies) for prohibitively expensive crushing, washing and drying facilities. In the balance of this statement I refer to the recognition of the matters that I have described in this paragraph as the Hypothesis.

32 Mr Vorias confirmed that the applicant embarked upon a series of activities to test that hypothesis. He explained:

> [I]n order to test the Hypothesis there would be a need for a bulk coal sampling program in order to undertake research and testing into the potential for new or improved mining methods and processing plant requirements. This would require survey work as well as some drilling and thereafter the excavation of test pits to enable bulk samples to be taken for the purpose of conducting experiments that would be directed to verifying or disproving the Hypothesis. It was important to me at this stage that the surveys and drilling could be undertaken as efficiently and on as

> limited a basis as possible, as the main purpose of them at this stage, was to help in identifying the best locations within the EPC area at which the test pits could be excavated. A further purpose of undertaking these activities, was to test whether the survey methodologies, which were new, were able to provide accurate results. It was necessary in this respect to compare the results of survey data against drilling results. In addition, I was aware that the results of these works might assist in developing a profile of the ground which would assist in developing coal extraction methods that were better able to target and extract coal, while extracting less overburden and waste than would ordinarily be extracted by large-scale open cut mining methods …

33 Mr Vorias also stated that in August 2011, the 2D seismic survey was undertaken using technology known as onSEIS. He said the seismic survey technology was only of limited effectiveness. It indicated the coal seams were at different depths underground from the depths known to be accurate from drilling results. He claimed the results of this work were not and could not have been known in advance on the basis of current knowledge.

34 Between August and October 2011, the SkyTEM surveys were carried out. Mr Vorias said he was informed the use of SkyTEM technology would enable predictions to be made as to the depth of weathering. In turn, this would enable predictions to be made as to whether coal would be "freshest" at a shallow depth. Mr Vorias stated that, from his perspective, the main purpose of the SkyTEM survey was to obtain data that would assist in determining where to excavate the test pits. However, a further purpose was to try the effectiveness of the technology, which prior to 2011 had not previously been used in the coal industry.

35 Mr Vorias said the main purpose of the drilling work at that stage, as well as the survey work, was to identify the optimal locations for the test pits/costeans. He explained samples from the drill cores were provided to A&B Mylec, a leading expert in coal preparation in Australia, for them to conduct washability testing. He anticipated that if the results were positive, larger bulk samples would have been provided for larger scale tests. In regard to the bulk samples, Mr Vorias accepted at the hearing before the Tribunal that the washability of the coal was a key criteria going to the quality of the coal and that the ease of handling – known in the trade as "handleability" – went to the economic viability of transporting the coal to market.

36 Mr Vorias hoped the tests undertaken by A&B Mylec would demonstrate a viable coking coal product, with acceptable levels of ash impurities, could be produced from the FCCM. Unfortunately, the testing done by A&B Mylec predicted unfavourable outcomes. As a result, Sedgman was engaged to review the procedures A&B Mylec had adopted and comment upon their processes and results. Ultimately, when their final reports were provided in June 2012, the A&B Mylec and Sedgman work confirmed the coal was unviable and the

hypothesis was disproved. The results suggested crushing and washing was unable to release enough ash to guarantee a quality that would be in accordance with internationally-traded coking product ash levels. Mr Vorias believed at this point that the applicant needed to reassess its approach and devise a new approach to manage:

(a) the nature and quality of the overburden;

(b) the depositional nature of the target coal seams;

(c) the washability of the product; and

(d) the eventual handleability of the coal.

37 Mr Vorias claimed in his evidence before the Tribunal that although the results of A&B Mylec and Sedgman were consistent with conclusions reached from early work done in relation to the FCCM, the outcomes of the work undertaken were not known or able to be known in advance.

38 Following receipt of the disappointing results confirmed by the reports from A&B Mylec and Sedgman, the applicant began to investigate alternatives. One of the possibilities considered was whether it might be feasible to process the coal by crushing it into very fine particles and undertaking further processing to remove impurities and then combining the coal particles into briquettes, which would be acceptable as meeting coking coal standards with respect to ash levels.

39 On 2 July 2012, it was agreed samples from the Wilton Project would be provided to Barrow Resources Pty Ltd (**Barrow**) for testing. The conclusions reached by Barrow supported the view that a viable coking coal product could be produced, but Barrow was concerned about whether the results of the laboratory testing could credibly be extended to justify the conclusions they had reached and whether their proposals could be used to develop a process that could be practically implemented. Mr Vorias said (in his evidence before the Tribunal) that the uncertainty over whether the results obtained by Barrow could be replicated on a larger scale involving the other coal deposits in the Wilton Project ultimately resulted in the applicant's board deciding not to proceed further with Barrow's proposals. Mr Vorias said that a number of other possibilities were considered including dry processing, but none of those options were considered to be viable. The possibility of briquetting was also not pursued.

40 On 12 October 2012, at Mr Vorias's request, Taggart provided a proposal to evaluate the coal washing characteristics and the potential for suitable liberation techniques to be developed. Mr Vorias said he instructed Taggart to proceed with the work.

41 Over the period December 2012 to January 2013, once the locations had been identified and the necessary approvals and land access obtained, the applicant began excavation of two costeans which targeted different points in the Wilton Project lease area. The samples from the costeans and drilling were delivered to Taggart and Virginia Tech for testing in early 2013.

42 On or around 24 January 2014, a Liberation Study Report was delivered by Taggart and provided to the applicant. Mr Vorias said that from the results of the testing carried out by Virginia Tech, Taggart were able to design a wash plant system that would be able to optimise the yield of coking coal. He said these results indicated that it may be possible to develop a process that would make mining the FCCM viable for the production of coking coal and that this was a very significant advancement in knowledge and was not something that was known in advance. In his oral evidence at the hearing before the Tribunal, Mr Vorias explained that it was the work that Virginia Tech undertook to identify an acceptable "crushing point" of the coal which was the significant discovery and "major win". Taggart then designed a wash plant that was able to wash the coal at that level. In his statement dated 3 August 2018, Mr Vorias stated that a patent application was lodged to protect the results of the experimental work performed by Virginia Tech on behalf the applicant. He stated that the results of the work done by Virginia Tech created a process for liberating coal that was markedly different from standard coal liberation processes.

Dr Vince's evidence

43 As noted by the Tribunal at [48], Dr Vince holds bachelor and doctoral degrees in chemical engineering from the University of Exeter, United Kingdom, and is a Chartered Engineer as well as a Fellow of the Institute of Chemical Engineers. He has over 33 years of domestic and international industrial experience in the coal industry, including nearly 20 years with BHP, with the last seven years as Superintendent Coal Processing Improvement with responsibilities for ten BHP coal preparation plants in Queensland, New South Wales and Indonesia.

44 In his written expert report dated 26 March 2018, Dr Vince provided an overview of his understanding of the applicant's registered core activities and registered supporting activity. He stated:

> 1.9 The difficulty with the FCCM measures has been generally and widely known in the public domain for many years … and is generally appreciated by Mr Vorias.
>
> …
>
> 1.12 **It is my opinion that this set of activities form part and parcel of normal mining and mine development activities. That is, they are, in my opinion, business-as-usual activities for a coal mining company attempting to make a business from a new tenement acquisition**. They are not, in my opinion, activities that may be regarded as or related to scientific R&D. This is primarily because the Scientific Method was not, in my opinion followed. This is evidenced by the overt absence of properly formulated hypothesis statements and proactive, interactive project or activity plans.
>
> 1.13 The activities I have examined are, in my opinion, **a set of loosely associated practical events whose purpose was to acquire measurement data. There was no R&D Plan followed that I could identify associated with any of the [applicant's] activities I examined.**
>
> 1.14 It is my opinion that the activities undertaken by [the applicant] comprised making measurements of important quantities and not scientific experiments. Such measurements are very important for the development of the mine, but did not, in my opinion, constitute scientific experiments.
>
> (Emphasis added by the Tribunal.)

45 Dr Vince outlined the numerous hypothesis statements contained in the applicant's evidence in the preamble to his report. He detailed, based on his experience, what he considered to be a formulation of a workable hypothesis statement in an R&D context. He described what comprised a scientific experiment and explained it would be centred on a well-defined hypothesis and progressed through an R&D plan. He opined that a "hypothesis statement is an initial building block in the Scientific Method upon which all R&D need to be soundly based". In regard to Mr Vorias's hypothesis contained in his statement dated 20 October 2017 at [37], Dr Vince opined it did not meet this requirement as it is "extraordinarily long, imprecise, ill-defined and is so nebulous and all-encompassing that it is difficult … to recognise it as a meaningful scientific hypothesis statement". In regard to the other hypothesis statements identified by Dr Vince in his report, he opined that they were all equally vague and were not scientific hypothesis statements for the current scientific R&D context. Rather, they were all statements of aims and objectives of the relevant activities. Dr Vince also opined that an R&D plan should be clearly documented prior to undertaking activities, should be forward-looking and expressed in the future tense.

46 In relation to Activity 1.1, Dr Vince expressed the following views:

> (b) … [T]he seismic and aerial survey activities will provide data that better define the coal measures but, pre-existing historical data relating to [the applicant's] Wilton Project tenement, and neighbouring properties, will allow very reasonable predictions to be made about coal quality and its probable distribution within the measures.
>
> (c) The actual extent of the better definition the surveys provide will not be known until all data are received, interpreted and reported. This, in my opinion, is the conventional way the coal mining business manages and controls the acquisition of important information …
>
> (d) In the case of recovery of bore core samples, the data collected therefrom will, in my opinion, very likely also be consistent with historical records and result only in an improved level of characterisation detail. This is because ... coal is laid out in seams in the ground and, in the main, presents in predictable layers.
>
> …
>
> (g) … [T]he outcome of Activity 1.1 could in general terms have been predicted and determined by appropriately qualified personnel in advance.
>
> (h) The uncertainty, in my opinion, will be in the detail of the characteristics of the coal measures that will be determined following routine analysis and measurement procedures. In addition, a carefully designed bore core drilling strategy will lead to minimal risks in the efficacy of the data determined.
>
> (i) In the case of small scale mining operations, correct selection and use of commercially available equipment by appropriate professional personnel will make it almost certain that the activity will be undertaken correctly. Whilst the exact nature of that outcome will not be known prior to conducting it, the general outcome is readily predictable, and the detailed outcome will add to the body of information and data to be considered in the mine development plan.

47 In his report, Dr Vince discussed in detail the activities undertaken by the applicant in relation to Activity 1.1. He opined that the surveys of the resource using SkyTEM and onSEIS technologies involved the application of relatively new but existing technology to the normal mining business activity of resource exploration with a view to turning the resource into a reserve. He stated he did not see an R&D plan driving the activities, nor any mention of a hypothesis being tested. Dr Vince saw the analysis of the data collected by the SkyTEM technology as a conventional activity to determine a workable understanding of the resource geology and as part of conventional mine planning activities. In regard to the bore core samples collected and analysed, Dr Vince stated this occurred using conventional coal characterisation methodologies. He did not regard the activities as "experimental activities whose outcome could not have been known or determined in advance, or whose outcome could only be determined by applying a systematic progression of work that proceeds from

hypothesis to experiment, observation and evaluation". He stated his opinion was based on the "distinct lack of an associated hypothesis statement and because the samples were analysed in routine ways by commercial techniques following conventional protocols".

48 Dr Vince opined that the activities could not be regarded as activities that were conducted for the purposes of generating new knowledge because "the practical tasks generated new data only on coal measures that have been known for many years". In his opinion, this was "useful to add to the already considerable databank relating to the FCCM, but did not generate any new knowledge". He said the 2D seismic survey, aerial survey, drilling and costean planning activities were clearly related to discovering deposits, determining the location of deposits and determining the size and quality of deposits. He stated that all of these activities are routine aspects of exploration and mine development to change a coal deposit classification from a resource to a reserve. He said that, in his mind, these activities were a routine part of exploration and mine planning and could not be considered as R&D.

49 In relation to Activity 1.2, Dr Vince stated:

> In my opinion, an appropriately qualified professional would be expected to know that the highly banded nature of the coal measures would cause poor yields, but that the exact value of the expected yield would have to be determined by analysing and measuring samples of the coal. The analysis procedures to be adopted for [the applicant's] mine planning and coal preparation activities were, in my opinion, routine and common to all coal mines around the world.
>
> It is also my opinion that an appropriately qualified professional would anticipate yield increases to be achieved by initiating liberation via crushing the coal …
>
> Once again, the exact effect of crushing on the extent of liberation would need be determined by conducting practical characteristic measurement activities. The procedures that should be adopted for such characterisations are routine and readily available at commercially engageable contract facilities …

50 In his report, Dr Vince stated that the issue report compiled by A&B Mylec was reflective of routine coal analysis. He said he could not identify a scientifically formulated hypothesis statement for the activity, nor could he identify a properly formulated R&D plan linking the activity to the wider activities. Dr Vince confirmed the second opinion report by Sedgman did not alter the nature of the original A&B Mylec report functions. As to the activities related to the examination of a beneficiation process, specifically the design of a wash plant, Dr Vince opined that such activities are routinely associated with any coal mine and coal beneficiation plant development.

51 In regard to Activity 1.3, Dr Vince stated:

> The outcome of liberating the FCCM by crushing would be the generation of ultra-fine coal particles and the subsequent need for ultra-fine coal processing techniques. I would expect this to be anticipated by appropriately qualified personnel and also that the options available to effect efficient beneficiation processing to be understood.
>
> This is because the methods for beneficiating fine coal particles are commonplace in the contemporaneous industry … and the extension to the ultra-fine sizes is a natural progression on the theme that also has been achieved at an industrial scale for many years …
>
> Therefore, it is my opinion that the broad outcome of the Core Activity 1.3 relating to beneficiating ultra-fine coal could be known or determined in advance of undertaking the activity. The fine details would be ascertained by limited well focused practical measurement activities. Such activities I would not regard as experimentation, but routine practical activities to determine variables to be used as inputs to established design processes.
>
> Once ultra-fine coal has been beneficiated, it is necessary to enlarge the size of the particle matrix to enhance the marketability of the product. The approach chosen to do this was the well-known tried and tested technology of briquetting, and since this is a well-known technology in commercial application … it requires only a limited amount of practical measurement activities. Such activity is routine and does not involve experimentation. It requires practical work to elaborate on the details of the final design.

52 In his report, Dr Vince noted samples were sent to Barrow for testing and analysis. He pointed out there was no linkage to an R&D plan or a scientific hypothesis statement. He stated he did not regard this activity as experimental, but as "routine practical test work to determine variables to be used as inputs to established design processes".

53 In regard to each of the claimed "core R&D activities" registered in the 2012 year, Dr Vince stated the applicant had not followed a systematic progression of work and opined the applicant "followed and applied pre-existing processes" such that there was no need for them to follow a systematic process to achieve their outcomes. He was of the opinion that the applicant applied "an ad-hoc approach, cherry-picking activities as and when they were thought of, because the outcomes were focused on acquiring data and making measurements as would typically be expected in due diligence-type investigations". Dr Vince said in summary:

> 1.27 If the Scientific Method cannot or has not been applied to a particular activity, then it is my strong opinion that such an activity cannot be considered an R&D activity from a scientific or engineering perspective.
>
> 1.28 The main reason for me forming the opinion that [the applicant] has not followed the conventional Scientific Method is that I could not find within the documentation provided to me a scientifically formulated hypothesis statement or scientifically formulated hypotheses statements.
>
> 1.29 I could not identify a single scientifically formulated hypothesis statement

> that I could associate with [the applicant's] activities. I have identified a number of statements that purport to be hypothesis statements in the material that I have considered in this matter but, for the reasons I have described in the main body of this report, they do not conform to my understanding of a scientifically formulated hypothesis statement. They all are, in my opinion, statements of the objective of the activity and do not include any testable component to the purported hypothesis statements. These purported hypothesis statements are, in my opinion, statements relating to the hopes and aspirations of the activity and are akin to those I would … expect to be included in a project proposal.

54 In summary, Dr Vince did not regard the registered core activities and registered supporting activity as experimental, but rather as "routine practical measurements to determine variables to be used as inputs to established design processes". He was of the opinion that the activities were conducted following standard approaches in the mining business, but he could not identify a "proactively developed and monitored R&D Plan encompassing a scientifically formulated hypothesis for activities relating them to the overarching project".

55 In regard to the systematic progression of work, Dr Vince said in his report that there were individual tasks that were carried out for or by the applicant that showed a steady progress towards an overall objective. However, the lack of an R&D plan and scientific hypothesis statement meant these activities were not conducted in a systematic way. He said there were individual groupings of activities that within themselves followed routine analysis procedures such as testing for washability or liberation, but there was no proactive R&D plan covering their coordination.

56 In regard to "existing knowledge or experience", Dr Vince stated in his report that, for him:

> [N]ew knowledge would represent a major departure from the current level of understanding or a discovery of something new. I would be looking for a step-change in our level of understanding, but what I see is a continuous incremental increase in the data and information that adds incrementally to the pre-existing body of knowledge.

57 Dr Vince addressed each registered core activity and the registered supporting activity in his report. In summary, he said:

> 1.18 … in general broad terms the outcome of the activities could have been predicted using historical and experiential evidence and data. In my opinion the predictions would be sufficient to be used for a rough examination of the phenomena involved, and so in that context, it is my opinion that the outcome could have been known in advance. Such predictions would give good broad and sufficiently precise estimates of the values of parameters and information design in the given situations.
>
> 1.19 This is particularly the situation in the coal mining business which is very mature and has the luxury of many decades of characterisation data to draw

> upon. However, defining specific details would require practical characterisation and measurement activities to be conducted.

58 Additionally, in regard to each of the claimed "core R&D activities" registered in the 2012 year, Dr Vince stated:

> In general, much data has been generated but there are no clear indicators that cite that new knowledge was generated from the activities or that the data has been converted to knowledge. Significant data was generated relating to better defining the resource and characterising the coal therein, but in my opinion, this does not constitute knowledge, it is just information and data to be used for commercial purposes…
>
> … These activities, in my opinion, generated data for mine and processing plant design purposes only, and that, in my opinion, does not constitute new knowledge.

Dr Euston's evidence

59 The Tribunal noted at [65] that Dr Euston holds a first class honours degree in mineral processing and a PhD in fine coal processing. He has worked in the Australian coal industry since 1985 and has held positions in research and development, process engineering, plant design and equipment supply.

60 In his report dated 20 December 2017, Dr Euston stated that the 2D seismic survey was performed to "discover the extent and distribution of the coal within the FCCM" and would be expected to produce "accurate information on the size, variability and quality of the coal deposits". He said the use of SkyTEM for evaluating coal deposits was a "novel application of this method of surveying" and that the drilling associated with the surveys "is an accepted component of the evaluation of any mineral or coal deposit" and was "required to accurately assess the size, quality and variability of the FCCM and to assist in the location of the most representative costean sites". Similarly, he said the costean planning and excavation was a required component of the detailed resource model. Dr Euston expressed the views that:

> [28] … For all coal and mineral deposits, seismic and other surveys and the drilling of test holes can contribute to the ultimate generation of a physical model of the coal or mineral deposit. These tests result in an accumulation of knowledge about the deposit, necessary to carry out all further activities such an economic evaluation of the deposit and ultimately, mine planning …
>
> [30] A core component of the hypothesis is that selective mining methods could be used to remove only the fresh coal and leave the weathered coal behind. In my opinion the four activities described above, (i) to (iv) in the list, together form a logical set of experiments and activities necessary to develop an accurate knowledge of the deposit and in turn to enable the hypothesis to be validated or disproved. The sequence of activities with appropriate reviews between them have followed the scientific method …

61 In regard to the A&B Mylec and Sedgman work, Dr Euston stated:

> [32] … A&BMylec have around 30 years' experience in coal testing and analysis and would be very familiar with the Bowen Basin coal deposits. I would assume that with this experience **they would have anticipated a less than favourable outcome from their work with the FCCM. The quality of these coals is well documented, and I am confident that A&BMylec would have been aware of this work**. However, the use of an experiment "to demonstrate a known fact" is an accepted purpose for an experiment. It was a logical input to the testing of the hypothesis to confirm that conventional coal preparation methods would probably not be effective. In this case, the activities leading up to the collection of the costean samples ensured that a representative sample of fresh coal was to be tested by A&BMylec, something which previous, less rigorous sampling and testing had not achieved.
>
> [33] … The work carried out by A&BMylec and subsequently confirmed by Sedgman (Activity (v)) closely followed the standard and accepted methods for coal testing and analysis. **The work to that stage had focused on determining the size and quality of the FCCM coal deposit. This work was aimed at determining how the coal is to be mined and the expected variability in quality** …
>
> [49] … A&BMylec would more than likely have been aware of the likely outcome of their work. As such the results could have been predicted, but not accurately known, in advance. As a result, I believe that this was a necessary step in the experimental process. Prior to embarking on subsequent activities, it was appropriate to formally discount the **conventional thinking**.
>
> (Emphasis added by Tribunal.)

62 Dr Euston said the outcomes of the work performed by Taggart and Virginia Tech could not have been known in advance. He said this was a "pivotal point in the project timeline" when "[c]onventional thinking has been replaced by new concepts". He described the Taggart and Virginia Tech work as going to the next stage of detailed testing and that it represented "a return to the first principles of coal and mineral beneficiation". In his report dated 20 December 2017, Dr Euston formulated his own project summary/plan of the activities performed by Taggart and Virginia Tech. That summary or plan included a hypothesis, a description of the experiment, observation/evaluation and conclusion.

63 In his statement dated 12 April 2018, Dr Euston accepted Mr Vorias's hypothesis statement and opined that there was an overriding hypothesis and concept for the Project and for the individual activities. Dr Euston opined that an essential component of the scientific method was the inclusion of a number of feedback loops. He explained:

> … at one or more stages the researcher is required to question the outcome or the value of the experiment. If necessary, the research [sic] can return to the hypothesis and adjust the plan to conduct further experiments.

64 Dr Euston went on to explain:

> [a] formal R&D plan is not necessary for the work to be described as systematic progression of work based on principles of established science that proceeded from hypothesis to experiment, observation and evaluation, leading to logical conclusions.

65 Dr Euston added Dr Vince's concept of an R&D plan was "idealistic and not what would normally occur in the course of an industry-based research program".

66 During the Tribunal hearing, Dr Euston accepted the initial surveys and the costeans were all done as part of the exploration of the coal resources at FCCM. He stated it was "part of the process" to ensure that when large bulk samples were taken, the sample is representative of what could be mined in much larger quantities during commercial operations.

Other evidence

67 At the hearing before the Tribunal, the applicant tendered the witness statements of Tom O'Malley and Kate Hine. These statements were admitted into evidence without objection and the witnesses were not required for cross examination.

68 In addition, the applicant tendered the witness statements of Dr Tobias Krull and Dr Peter Bethell. These witnesses were not required for cross examination. The Tribunal gave those witness statements limited weight due to relevance; both related to activities conducted outside of the 2012 year – namely, the work conducted by TUNRA in 2015 and the work conducted by Virginia Tech and Taggart.

The registered activities

69 On 30 April 2013, the applicant applied to register four activities as core R&D activities for the 2012 year, namely:

(a) Design of selective mining process (referred to as Activity 1.1);

(b) Design of wash-plant beneficiation process (referred to as Activity 1.2);

(c) Design of ultra-fine processing and briquetting (referred to as Activity 1.3); and

(d) Design of the geochemistry model for environmental waste management (referred to as Activity 1.4).

70 As noted in the Reasons at [30], the applicant did not press its claim with respect to Activity 1.4. That activity can therefore be put to one side.

71 The applicant also applied to register one supporting activity, referred to as Activity 1.1.1. This was titled "Background research" and was claimed to be directly related to Activity 1.1.

72 The applicant identified that the core activities and the supporting activity were to be undertaken as part of the Project from July 2011 to June 2014.

73 The application form contained the following descriptions of the Project and of activities 1.1, 1.1.1, 1.2 and 1.3 (omitting fields that are not presently relevant):

> **Project title* Design and development of a new mining and beneficiation process**
>
> **Objectives of the project***
>
> Introduction:
>
> In coal mining, the key consideration is to use the most appropriate mining and beneficiation process to ensure financial viability; i.e. a combination of technique(s) used to extract the resource from the ground and the process(es) required to obtain a commercially viable product from the run-of-mine (ROM) coal.
>
> For most large, coa1 mining companies, the mining process invo1ves identifying low-density coal seams; using large-scale bulk mining techniques; and then coal preparation in a traditional, dense-medium Coal Handling and Preparation Plant (CHPP), which involves extracting viable coal product from ROM coal that has few impurities (i.e. high-density, non-coal material).
>
> In current Queensland coal mining operations, impurities are made largely of loose mine-roof and floor material, the liberation process – separation of coal from non-coal material – is often a simple process.
>
> Comparatively, coal seams, such as the Fort Cooper Coal Measures (FCCM), are more difficult and less economical to mine.
>
> Problem:
>
> The quality of coal that can be extracted from both types of coal seams is dependent on a number of geological factors, specifically relating to individual seams. The problem is that the cost of extracting viable coal from the FCCM is much higher. Not only is it more difficult to extract coal from banded coal seams, the inherently high ash content of the coal seam means:
>
> • Higher ash range of coal – To produce coal suitable for the coking coal market, it has to be at a certain ash range. Ash in coking coal creates a reject material called slag, which reduces the efficiency of blast furnaces. Consumers gain no benefit from buying ash. The higher the ash content, the lower the price of the coal.
>
> • Lower yield – Yield is a factor of product obtained to waste remaining from beneficiated ROM coal. Due to the high ash content of the coal, a different liberation process has to be adopted, which may require crushing the coal to fine-sized particles that may be too small to process or capture after washing.
>
> To cope with increasing demand for coal products, modern mining methods that make the mining and beneficiation of banded, inherently high-ash coal seams practical and economically viable must be developed.

Solution:

Problems associated with mining banded coal seams are largely due to the lack of technology for mining, handling and preparing coal from banded and atypical coal seams.

Specifically, this project examines the deficiencies of existing coal mining processes for the mining of banded coal seams and the handling of high-ash coal seams within the FCCM in Queensland. The main challenge of this project is the beneficiation, which is a worldwide problem that scientists are trying to overcome in the development of a new generation of coal deposits and coal mining process.

The hypothesis is that improvements to the existing coal mining and beneficiation process must be made to obtain an economically viable product from the coal seams within the FCCM. These improvements relate specifically to the combination of:

• Developing a selective mining process that can accurately define the 'coal rich' bands from the 'waste rich' bands;

• Identifying an optimum liberation process and particle size for the inherently high-ash coal;

• Designing and developing a CHPP for the economic beneficiation of the ROM coal;

• Identifying an effective method to capture ultra-fines as a high-quality, low-ash, hard coking coal product;

• Designing and developing a geochemical model that can be continually calibrated during the project's lifetime to ensure mined waste rock materials are managed to prevent the creation of long-term MMD legacy issues.

New knowledge*

The main item of new knowledge is in the form of a new mining and beneficiation process that can produce an economically viable product from the banded, inherently high-ash resource within the FCCM.

The mine that QCC is working on to facilitate this development is the first stand-alone FCCM mine in the Bowen Basin. The success of this new mining and beneficiation process will unlock a significant amount of previously unviable coal resources to produce viable high-grade coking coal that can help with meeting the increased worldwide demand for coal and, consequently, contributing to building the Queensland and Australian economy.

Specific items of new knowledge generated from undertaking the project are listed below. They pertain to overcoming the knowledge gap for the design and development of the new process:

• Selective mining strategy: Understanding of how use the SkyTEM technology to accurately define 'coal rich' banded sequences for selective mining.

The SkyTEM technology has been used on only one coal project before QCC. The technology was initially developed to model ground water in Denmark and involves measuring the electromagnetic induction of the ground through variations in water porosity, salinity, saturation, as well as clay content and weathering.

The data collected from the SkyTEM survey is used to develop a profile of the subsurface. The profile does not locate the coal, but illustrates the geological structures of the ground to allow geologists to more effectively target effective drill

locations.

During the drill process, the SkyTEM data is calibrated against the real data collected from the drill hole to confirm the structures and depth of weathering illustrated in the profile. From this process, an accurate model can be developed, assessing the variations in depth to mineable resources.

• Pre-treatment process: Understanding of the best methods for removing waste rock from feed material before it goes through the wash plant; in particular, determining the viability of 'dry' beneficiation techniques on ROM coal from the FCCM.

• Fine-circuit wash plant: Undemanding of how to design and develop a wash plant with fine circuit for the processing and capture of fine and ultra-fine coal particles.

Fine circuits exist as a technology in the treatment of traditional ROM coal. Traditional ROM coal is processed as larger-sized particles.

The majority of traditional ROM coal does not require processing within the fine circuit because they are inherently lower in ash content; therefore, traditional fine circuits are not designed to process and capture large quantities of fine and ultra-fine coal particles.

QCC is developing a fine circuit as a large-volume operation within the wash plant, so that a large percentage of their product can be obtained through the fine circuit.

• Ultra-fines and briquetting: Understanding of how to develop a lower-ash, higher-yield product from ultra-fines through the process of briquetting.

Briquetting as a technique has been applied in mines in Victoria and Western Australia to remove moisture from coal. QCC is attempting to use briquetting as a technique to capture fines and ultra-fines, in order to create a high-quality coking product.

• Geochemical modelling: Understanding of how to design and develop a geochemical model that can be continually calibrated during the project lifetime to ensure mined waste rock materials are managed to prevent creation long-term MMD legacy issues. The geochemical model will help QCC determine the most cost-effective waste management solution for their operations in the FCCM.

1.1 – Design of selective mining process

Core R&D activity description*

• Practical experimentation with new technology:

- 2D seismic survey acquisition to help define resources, using the revolutionising onshore and shallow water acquisition technology – onSEIS;
- SkyTEM electromagnetic survey to map the distribution of subsurface electrical conductivity via electromagnetic induction.

• Analyse data collected from the SkyTEM survey to identify the 3D voxel model of the base of weathering and seam subcrop geometries.

• Collect and analyse real data from:

- HQ/PQ Coring for coal quality and washability;
- LD cores to allow washability testing of coals to be undertaken. The LD cores provide data on the specific banded nature of the coal seams and, in conjunction

with the washability data, allow the definition of selective mining working sections;

- Chip holes to support/validate SkyTEM data.

• Complete two investigative mining operations, (the costeans) for the scaled implementation and testing of theories developed about the ability of current mining equipment and operators to be able to selectively mine with any degree of success within the FCCM.

• Refine strategies and techniques to establish selective mining process.

1.1.1 – Background research

Supporting activity description*

• Literature search and review of existing mining and beneficiation processes and technologies.

• Geographical survey via 2D seismic survey and SkyTEM electromagnetic survey to help define resources.

• Consultation with industry professionals and potential customers to determine the level of interest and commercial feasibility of such a project;

• Preliminary equipment and resources review with respect to capacity, performance and suitability for the project;

• Consultation with key component/part/assembly suppliers to determine the factors they consider important in the design, and to gain an understanding of how the design needs to be structured accordingly.

1.2 – Design of wash-plant beneficiation process

Core R&D activity description*

• Assessment of ROM coal quality to determine beneficiation strategies:

- Retrieval of samples for laboratory testing:

- HQ/PQ Coring for coal quality and washability;

- LD coring for CQ and washability;

- CQ testing carried out to allow the compilation of conceptual models, simulations, possible prototyping, etc.

• Coal quality testing and yield assessment.

• Examination of beneficiation processes to optimise economic potential:

- Pre-treatment options to reduce waste feed to plant

- Design of a large fine circuit for capturing fines and ultra-fines within the wash plant.

- Conceptual sketches;

- CAD/3D design simulation;

- Rapid prototyping;

- Review of design.

• Feasibility test and final design of wash plant by external contractor.

• Process beneficiation flow diagrams.

1.3 – Design of ultra-fine processing and briquetting

Core R&D activity description*

• Research and testing for ash content and yield.

• Research and testing for product handleability.

• Research and testing for process scalability.

74 These activities were registered by the Board for the 2012 year.

The initial decision and the internal review decision

75 On 28 May 2015, pursuant to s 27J of the IR&D Act, a delegate of the Board decided that none of the registered core activities, and the registered supporting activity, for the 2012 year were eligible to be considered as "core R&D activities" or as "supporting R&D activities" within the meaning of s 4 of the IR&D Act and ss 355-25 and 355-30 of the ITAA 1997. The applicant sought internal review of this decision.

76 On 27 January 2017, pursuant to s 30D of the IR&D Act, the Board made the internal review decision. This confirmed the decision of the delegate dated 28 May 2015.

The Tribunal's decision

77 On 30 January 2020, the Tribunal affirmed the internal review decision and provided the Reasons.

78 After setting out extracts from the applicant's application for registration for the 2012 year, the Tribunal at [30] of the Reasons summarised the activities registered as core R&D activities for the 2012 year that required evaluation as follows:

(a) the SkyTEM survey;

(b) the 2D seismic survey;

(c) drilling to validate the survey results and provide samples to A&B Mylec and Sedgman for analysis; and

(d) the A&B Mylec and Sedgman analysis.

On appeal, there is no issue that the Tribunal correctly summarised the applicant's activities requiring evaluation.

79 At [32]-[33] of the Reasons, the Tribunal identified the issues for determination as follows:

> 32. The substantive issues for determination in these proceedings are whether for the 2011-2012 year:
>
> - all or part of each of the registered core activities were "*core R&D activities*" within the meaning of that term in s 355-25 of the ITA Act; and
>
> - The registered supporting activity was a "*supporting R&D activity*" within the meaning of the term in s 355-30 of the ITA Act.
>
> 33. The applicant also contends in its Statement of Facts, Issues and Contentions that issues may arise as to whether:
>
> - part of any of the registered core activities found to be an activity referred to in s 355-25(2) of the ITA Act was for the purpose of paragraph 27J(1)(c)(iii) of the IRD Act, a "*supporting R&D activity*" conducted in the 2011-2012 year in relation to one or more specified core R&D activity (activities) yet to be conducted for which the entity could be registered in the 2011-2012 year if that activity was conducted in that year; and
>
> - part of any of the registered core activities found to be an activity referred to in subsection 355-25(2) of the ITA Act ought to be registered as a supporting activity being an activity undertaken in the 2011-2012 year for the dominant purpose of supporting one or more core activities.

In oral submissions on the appeal, senior counsel for the applicant submitted that the issues referred to in [33] of the Reasons were raised in *the Board's*, rather than the applicant's, statement of facts, issues and contentions. It was not suggested, however, that anything turns on this.

80 At [75]-[82], the Tribunal made some general observations about the application of the legislation, referring to the judgment of the Full Court in *Moreton*.

81 The Tribunal's core reasoning on the application of the relevant provisions to the activities requiring evaluation was at [83]-[92] of the Reasons.

82 At [84], the Tribunal reiterated the core activities in question in the Tribunal proceeding, as set out in [78] above.

83 At [85], the Tribunal stated that the 2D seismic survey "was performed to discover the extent and distribution of the coal within the FCCM to aid the production of accurate information on the size, variability and quality of the coal deposits". The Tribunal also stated that the "drilling activities were undertaken to validate the survey results".

84 The Tribunal made some further findings at [86]-[87]:

> 86. The evidence of Kate Hine, a geophysicist at Groundprobe Geophysics Pty Ltd (Groundprobe Geophysics) during the relevant time, is that the use of SkyTEM for evaluating coal deposits was a novel application for the method of surveying. Prior to 2011, the SkyTEM technology for coal projects was completely untested. Testing of SkyTEM's effectiveness as a survey technique in greenfield coal sites was the research project of Groundprobe Geophysics. The applicant agreed to participate in this trial. For the applicant's purposes, the results generated by the use of the SkyTEM technology enabled them to identify the depth of weathering within the FCCM, which in turn enabled them to predict whether the coal would be "freshest" at a shallow depth. As expected, this assisted the applicant in identifying the ideal location for drilling and the excavation of test pits. It is plain that although the use of the SkyTEM technology in the Wilton Project was a novel use of the technology, this aspect was not part of the applicant's activities but rather a research project conducted by Groundprobe Geophysics. The applicant's use of the results produced by the SkyTEM survey merely contributed to the accumulation of knowledge about the extent, location and quality of the coal within the FCCM. This is further supported by the evidence of Tom O'Malley, a geologist who participated in the conduct of the 2D seismic and SkyTEM surveys, who advised that the survey work enabled the applicant to "identify where the better areas of coal were located in the permit area".
>
> 87. It is generally well accepted that coal found in the FCCM has high ash content and is uneconomical to mine using current processes. Despite the available knowledge about the FCCM, the 2D seismic survey, SkyTEM aerial survey and drilling activities undertaken by the applicant enabled a more detailed and exhaustive exploration of the coal deposits in the FCCM. The activities enabled the applicant to discover the extent and distribution of the coal within the FCCM by mapping more precisely the location of the deposits including the depth, the depth of weathering, and the quality of those deposits. All of these activities could contribute to the ultimate generation of a physical model of the coal deposit in the FCCM and result in an accumulation of knowledge about the deposit. That would assist the applicant to identify ideal locations to obtain bulk samples for further testing. The work undertaken by A&B Mylec and Sedgman flowed from these processes.

85 The Tribunal then reasoned as follows at [88]-[92]:

> 88. The evidence shows the work undertaken by A&B Mylec was routine coal analysis in accordance with Australian Standards. A&B Mylec were engaged to undertake testing on bulk samples to better determine the washability and handleability of the coal deposits within the FCCM. The washability of the coal went directly to quality and the handleability provided an indication as to the ease with which the coal could be transported to market. Unsurprisingly, given the evidence, the results of the tests undertaken by A&B Mylec were unfavourable. Sedgman was engaged to confirm the results of A&B Mylec's work, which again, unsurprisingly, were unfavourable. **The experts are in agreement that the outcome of the work undertaken by A&B Mylec and Sedgman was predictable based on experience and the current knowledge available about the FCCM, could have been determined in advance and did not generate any new knowledge**.

89. The disappointing yet predictable results from A&B Mylec and Sedgman confirmed the coal was unviable. The results suggested that crushing and washing the coal was unable to release enough ash to guarantee a quality that would be in accordance with internationally traded coking product ash levels. According to Mr Vorias' evidence, it was only after these unfavourable results from A&B Mylec and Sedgman that the applicant reassessed its approach and devised a new approach to manage the nature and quality of the overburden; the depositional nature of the target coal seams; the washability of the product; and the eventual handleability of the coal. This new approach involved an investigation into whether it might be feasible to process the coal by crushing it into very fine particles and undertake further processing to remove impurities, evaluate the coal washing characteristics and develop suitable liberation techniques. This led to the Barrow, Taggart and Virginia Tech work. As Dr Euston explained, the Taggart and Virginia Tech work "*was a pivotal point in the project timeline when conventional thinking has been replaced by new concepts*". There was plainly a step change in the thinking after the A&B Mylec and Sedgman results such that a new project was instigated by the applicant.

90. The applicant predominantly relies on the hypothesis statement of Mr Vorias (and numerous iterations in similar terms) to shed light on the ultimate purpose and basis for the activities registered and undertaken by the applicant in the 2011-2012 year. Having carefully considered the outcome of the activities, it is apparent they do not marry well with the hypothesis statement. **While the activities appear to involve a logical progression of activities to acquire knowledge about the FCCM, there are no R&D plans or documentation to demonstrate the activities were carried out by applying a systematic progression of work based on principles of established science; or that they proceeded from the purported hypothesis to experiment, observation and evaluation, leading to logical conclusions**: see *Rix's Creek Pty Limited; Bloomfield Collieries Pty Limited and Innovation Australia* [2017] AATA 645; *Docklands Science Park Pty Ltd and Innovation Australia* [2015] AATA 973. **Nor were the activities in question conducted for the purpose of generating new knowledge**. As Dr Euston characterised the activities, they merely built on the body of knowledge already available about the FCCM. The activities were appropriate in so far as they formally checked the conventional thinking before other activities and concepts could be pursued. It was only after the disappointing yet predicable results of A&B Mylec and Sedgman that the applicant turned its mind to new concepts to identify improvements to be made to existing coal mining and beneficiation processes, or new processes which could be developed to enable the mining, processing and handling (transporting) of the coal in the highly banded coal seams in the FCCM. It may be that those activities conducted by the applicant could be said to be in line with the purported hypothesis statement. However those activities fall outside of the 2011-2012 income year.

91. We are satisfied the activities undertaken by the applicant up to and including the A&B Mylec and Sedgman results were focused on determining the size and quality of the FCCM coal deposit and were aimed at determining more precisely the expected variability in quality. They were all generic exploration activities undertaken in the initial exploration stages which a company with a mining tenement would undertake in order to ascertain the location, quality and size of the coal resources so it can progress to a point of being able to mine the coal. **Accordingly, we find the core activities**

> **registered and conducted during the 2011-2012 year, namely the 2D seismic survey, the SkyTEM survey, the drilling to validate the survey results and provide samples to A&B Mylec and Sedgman for analysis, and the A&B Mylec and Sedgman analysis, were not core R&D activities and all come within the exception in s 355-25(2)(b) of the ITA Act**.
>
> 92. The applicant contends in the alternative that the drilling and survey activities were "*supporting R&D activities*" because they were conducted for the dominant purpose of supporting a core R&D activity, namely, the excavation of costeans. It is accepted that the preparation for the location of the test pits may have begun in the 2011-2012 year and that the survey and drilling activities lead logically to this next step, however, we consider these steps were taken in the course of exploration within the FCCM and therefore come within the exemption in s 355-25(2)(b) of the ITA Act. The excavation of the costeans was undertaken in late December 2012/January 2013. The material dug from the costeans was sent to Taggart and Virginia Tech for detailed coal quality assessment. There is no evidence supporting a finding that digging the costeans satisfies any of the requirements of a core R&D activity nor does the ultimate use of the costeans impart to the activity that of an experimental activity within s 355-25(1). At hearing, Mr Vorias' agreed there was nothing particularly novel about the technology involved in the digging of a costean and that it was just to get a bulk sample. Dr Euston understood that "*costeaning refers to the practice of extracting small test pits to extract sufficient sample for analysis and testing*" and that it was "*one of the contributing factors to the body of knowledge required to fully determine the size and quality of a deposit*".
>
> (Emphasis added.)

86 Accordingly, the Tribunal affirmed the internal review decision.

The appeal on a question of law

87 The applicant appeals to this Court pursuant to s 44 of the AAT Act. An appeal pursuant to that provision must be "on a question of law": see generally *Haritos v Federal Commissioner of Taxation* (2015) 233 FCR 315 at [192]-[202].

88 The applicant's amended notice of appeal contains 12 "questions of law" (including a question numbered 1.1) and 12 corresponding grounds of appeal. In the applicant's amended outline of submissions, the applicant identifies five issues that arise on the appeal, as follows:

(a) The first issue – to which grounds of appeal 1, 1.1 and 2 are directed – is whether the Tribunal at [88] of the Reasons made findings that were not open on the evidence and applied a wrong legal standard, in considering whether the applicant's activities were of a kind "whose outcome cannot be known or determined in advance" within the meaning of s 355-25(1)(a) of the ITAA 1997.

(b) The second issue – to which ground of appeal 3 is directed – is whether the Tribunal erroneously substituted an alternative inquiry for that mandated by the statutory

language in s 355-25(1)(a), which requires consideration of whether the outcome of activities "can only be determined by applying a systematic progression of work that: (i) is based on principles of established science; and (ii) proceeds from hypothesis to experiment, observation and evaluation, and leads to logical conclusions".

(c) The third issue – to which grounds of appeal 5 and 6 are directed – is whether the Tribunal at [90] of the Reasons made findings that were not open on the evidence and applied a wrong legal standard, in considering whether the applicant's activities were "conducted for the purpose of generating new knowledge (including new knowledge in the form of new or improved materials, products, devices, processes or services)" within the meaning of s 355-25(1)(b).

(d) The fourth issue – to which grounds 6, 7 and 8 are directed – is whether the Tribunal misapplied s 355-25(1) by assessing the activities predominantly by reference to those undertaken in the 2012 year as opposed to taking into account the applicant's overall project, which would inevitably take several years, and consequently approached the application of s 355-25(2)(b) from an invalid starting point.

(e) The fifth issue – to which grounds 9 to 11 are directed – is whether the Tribunal erred by ignoring relevant evidence, and whether it failed to give adequate reasons to the extent that it did not accept the applicant's evidence, and whether it failed to make findings on material questions of fact.

89 By its proposed further amended notice of appeal (second version), the applicant seeks to raise an additional "question of law" (numbered 12) and two associated grounds of appeal (numbered 8.1 and 8.2). In summary, the applicant contends that the Tribunal erred by failing to consider a submission advanced by the applicant below to the effect that activities undertaken in the 2012 year should be regarded as "supporting R&D activities" on the basis that they were directly related to and carried out for the dominant purpose of supporting core R&D activities in later years, namely the work by Taggart and Virginia Tech. While the Board opposed the initial version of the proposed further amended notice of appeal, on the basis that the additional paragraphs used the word "including" in relation to the core R&D activities in later years, and thus did not define those activities with specificity, the Board did not oppose the amendments if the word "including" were replaced with "namely". The applicant responded that it was prepared to replace the word "including" with "namely" and subsequently provided the second version of the proposed further amended notice of appeal, which includes this amendment. In these circumstances, it is appropriate to grant the

applicant leave to amend its amended notice of appeal to the form of the proposed further amended notice of appeal (second version).

Consideration

90 We will consider each of the issues set out in [88] above in turn, and then consider question of law 12 and the associated grounds.

The first issue

91 The first issue is whether the Tribunal (at [88] of the Reasons) made findings that were not open on the evidence and applied a wrong legal standard, in considering whether the applicant's activities were of a kind "whose outcome cannot be known or determined in advance" within the meaning of s 355-25(1)(a) of the ITAA 1997.

92 The applicant's submissions focus on the final sentence of [88] of the Reasons (see [85] above), in which the Tribunal concluded that the experts were in agreement that the outcome of the work undertaken by A&B Mylec and Sedgman: (a) was predictable based on experience and the current knowledge available about the FCCM; (b) could have been determined in advance; and (c) did not generate any new knowledge. As to (a), the applicant submits that the opening words of s 355-25(1)(a) require consideration of whether the outcome of the activity in question "cannot be known or determined in advance"; the Tribunal wrongly equated the issue of whether an outcome can be known or determined with whether it can be predicted. As to (b), the applicant submits that, as a matter of fact, the experts were not agreed that the outcome of the work referred to at [88] "could have been determined in advance" and it was not open to the Tribunal so to conclude. As to (c), the applicant submits: first, as a matter of fact, the experts were not agreed that the outcome of the work referred to at [88] did not generate any new knowledge and it was not open to the Tribunal so to conclude; secondly, in considering whether the applicant's activities did or did not in fact generate new knowledge the Tribunal applied the wrong legal standard; the question s 355-25(1)(b) asks is whether the activities were conducted *for the purpose of* generating new knowledge.

93 The applicant's more detailed submissions on this issue can be summarised as follows:

(a) The Tribunal made critical erroneous findings with regard to the evidence relating to the work done by A&B Mylec and Sedgman. In general terms, that work sought to quantify the extent to which FCCM coal was liberated from impurities when crushed

to samples of different maximum sizes. It is incorrect, and it was not open to the Tribunal to conclude, either (a) that the experts were agreed that the outcome of the work undertaken by A&B Mylec and Sedgman could have been determined in advance and did not generate any new knowledge; or (b) that that was in fact the case. The Tribunal, which published its decision some 17½ months after the hearing, appears to have reached this erroneous conclusion by simply copying verbatim an assertion that had been made in the Board's closing written submissions (at paragraph 84), which cited only Dr Vince's report, and parts of it that did not, in any event, support what was asserted. It was pointed out in closing address that the Board's written submissions were wrong on this point, but the Tribunal failed to address this. Given the delay and the failure to address the applicant's argument, a reasonable inference or apprehension arises that the Tribunal was "unable to grapple adequately with the issues and, in the end, had become attracted to the decision which was the easiest to make" and that the Tribunal's "failure to deal in a significantly delayed judgment with particular matters on which the [applicant] relied in contradiction of the findings made … resulted from those matters being overlooked": cf *Monie v Commonwealth* (2005) 63 NSWLR 726 at [43]-[44]. The repetition of the Board's written submissions gives no indication that the applicant's contentions on this issue were given any independent consideration and in this respect the reasons below "do not contain indicia … that they are the product of the active application of an independent and impartial mind, which has absorbed and analysed the competing submissions": cf *Li v Attorney General (NSW)* (2019) 99 NSWLR 630 at [132].

(b) Moreover and contrary to the Tribunal's conclusion, the Board's expert, Dr Vince, accepted that nobody had previously done the work that A&B Mylec and Sedgman did (Tribunal transcript, day 5, p 288). Dr Vince gave evidence that it is not possible to quantify yield increases (i.e. recovery of coal after liberating ash and other impurities) without actual testing, and that one cannot estimate quantitatively the extent of liberation in the absence of testing.

(c) The applicant's expert, Dr Euston, gave evidence that the outcome of the work by A&B Mylec was experimental and that while they would have anticipated a less than favourable outcome from their work with the FCCM, the overall outcome of that work, which could have been predicted, could not have been accurately known in advance.

(d) The experts were not, therefore, in agreement that the outcome of the work undertaken by A&B Mylec and Sedgman could have been *known or determined* in advance. Further, the evidence of Mr Vorias was that: "The outcomes of the work undertaken by A&B Mylec and Sedgman were not known or able to be known in advance."

(e) The statutory requirement in s 355-25(1) that the outcome of work cannot be "known or determined" in advance requires a level of certainty that goes beyond whether or not an outcome can be predicted.

(f) The Tribunal therefore applied a wrong legal test in determining the issue with reference to whether the results were "predictable". It was also wrong, and irrelevant, to determine the issue adversely to the applicant by reference to whether or not the work in fact generated new knowledge: the statutory requirement is that the *purpose* of the activities is the generation of new knowledge. Whether new knowledge is generated or not is neither here nor there. The result *might* have been different if the Tribunal had not erred by making findings not open on the evidence and applying criteria other than those mandated by the statute: cf *Comptroller-General of Customs v Pharm-A-Care Laboratories Pty Ltd* (2020) 375 ALR 98 at [40].

94 During the hearing of the appeal, senior counsel for the applicant was asked by the Court whether the contention that certain findings were "not open" on the evidence was in substance a 'no evidence' ground. Initially, this question was answered affirmatively. However, subsequently, senior counsel said that the applicant did contend that the relevant findings were "not open". The difficulty with the latter proposition is that it is not self-evident that such a contention raises a question of law. If and to the extent that the applicant submits that the relevant findings were wrong, it is established that there is no error of law simply by making a wrong finding of fact: see *Minister for Immigration and Multicultural Affairs v Al-Miahi* (2001) 65 ALD 141 (***Al-Miahi***) at [34]. On the other hand, it is established that:

(a) whether a fact is supported by any evidence is a question of law (*Australian Broadcasting Tribunal v Bond* (1990) 170 CLR 321 (***Bond***) at 355-356; *Kostas v HIA Insurance Services Pty Ltd* (2010) 241 CLR 390 at [91]); and

(b) whether a particular inference can be drawn from facts found is a question of law; that is because, before an inference may be drawn, there is the preliminary question

whether the evidence reasonably admits of different conclusions (*Bond* at 355-356; *Al-Miahi* at [34]).

95 To the extent that the applicant seeks to invoke either or both of these propositions (as indicated in oral submissions at T13-T14), we accept that the submissions give rise to a question of law. However, as discussed later in these reasons, some of the applicant's submissions amount to no more than a submission that the Tribunal's findings were incorrect; such a submission does not give rise to a question of law.

96 For the reasons that follow, we do not accept the applicant's contentions in relation to the first issue.

97 Insofar as the applicant challenges the Tribunal's finding that the experts were agreed that the outcome of the work undertaken by A&B Mylec and Sedgman "could have been determined in advance and did not generate any new knowledge", there *was* evidence to support this finding, and the evidence *did* reasonably admit of this inference.

98 Dr Vince's evidence, as set out at [43]-[58] above, was to the effect that, while the *precise yield values* could not have been known in advance, the *outcome* of the work could have been known and determined in advance, and the work did not generate new knowledge (in the sense of a major departure from the current level of understanding or a discovery of something new). For example, in relation to Activity 1.1, Dr Vince opined that the outcome of the activity "could in general terms have been … determined by appropriately qualified personnel in advance" (see [46] above). In relation to Activity 1.2, Dr Vince's view was that "an appropriately qualified professional would be expected to know that the highly banded nature of the coal measures would cause poor yields, but that the exact value of the expected yield would have to be determined by analysing and measuring samples of the coal" (see [49] above). Further, Dr Vince expressed the opinion in relation to the registered activities generally that "the outcome could have been known in advance" and that "much data has been generated but there are no clear indicators that cite that new knowledge was generated from the activities or that the data has been converted to knowledge" (see [57]-[58] above).

99 Dr Euston's evidence also provided support for the Tribunal's findings. In particular, in regard to the A&B Mylec and Sedgman work, he stated (see [61] above):

(a) A&B Mylec "would have anticipated a less than favourable outcome from their work with the FCCM. The quality of these coals is well documented, and I am confident that A&BMylec would have been aware of this work".

(b) The work carried out by A&B Mylec and subsequently confirmed by Sedgman "closely followed the standard and accepted methods for coal testing and analysis. The work to that stage had focused on determining the size and quality of the FCCM coal deposit. This work was aimed at determining how the coal is to be mined and the expected variability in quality".

(c) A&B Mylec "would more than likely have been aware of the likely outcome of their work. As such the results could have been predicted, but not accurately known, in advance".

100 Thus, both Dr Vince and Dr Euston provided evidence that supported the Tribunal's findings that the outcome of the work undertaken by A&B Mylec and Sedgman "could have been determined in advance and did not generate any new knowledge". In the case of Dr Vince, he provided evidence in these terms. In the case of Dr Euston, the evidence referred to in [99] above, in the context of the evidence generally, supported the findings.

101 A theme running through the applicant's submissions is that the precise data produced by the tests conducted by A&B Mylec could not have been known in advance. The difficulty with this line of argument is that it fails to read s 355-25(1) in its entirety. The section refers to experimental activities "whose outcome cannot be known or determined in advance on the basis of current knowledge, information and experience …". During the hearing of the appeal, senior counsel for the Board provided an analogy of a person having a routine blood test for their cholesterol level. In that analogy, merely because the precise data or results cannot be known in advance does not mean that the test constitutes an experimental activity whose outcome cannot be known or determined in advance as referred to in the section. That analogy is useful in exposing the difficulty with the applicant's approach. In the present case, while the precise data produced by the work undertaken by A&B Mylec could not have been known in advance, it does not necessarily follow that the work constituted an activity whose outcome could not have been known or determined in advance on the basis of current knowledge, information and experience.

102 Insofar as the applicant submits that the Tribunal appears to have reached its conclusion by "simply copying verbatim an assertion that had been made in the respondent's closing written

submissions (at para [84])", we do not accept this submission in relation to the last sentence of [88] of the Reasons. The Tribunal's finding in that sentence is not in precisely the same terms as paragraph 84 of the Board's closing written submissions. That paragraph included: "Both Dr Vince and Dr Euston agree that the outcome of the work could have been determined in advance [fn 44]. It was not conducted for the substantial purpose of generating new knowledge." Further, the Tribunal had earlier set out, in some detail, the relevant portions of the evidence of Dr Vince and Dr Euston that supported its findings. In the circumstances, it was not necessary for the Tribunal to refer in its Reasons to the applicant's submission in closing address on this point, and the delay in delivering a decision does not provide a sufficient basis to infer that the Tribunal failed to have regard to that submission.

103 Insofar as the applicant submits that "contrary to the AAT's conclusion, the respondent's expert, Dr Vince, accepted that nobody had previously done the work that A&B Mylec and Sedgman did", citing Tribunal transcript, day 5, p 288, it is not apparent that the submission raises a question of law as distinct from a submission that the factual finding was wrong. In any event, the submission does not convey the full substance of Dr Vince's evidence in the passage cited. The full passage of the cross-examination of Dr Vince reads:

> [Counsel:] That work, nobody else had done that work in relation to the Fort Cooper coal until MILEC, Sedgman, Virginia Tech, have they?
>
> [Dr Vince:] That particular work, no, but the knowledge about its performance in those sort of processes was well known, and it doesn't - - -

104 Insofar as the applicant submits that Dr Vince gave evidence that it is not possible to quantify yield increases (i.e. recovery of coal after liberating ash and other impurities) without actual testing, and that one cannot estimate quantitatively the extent of liberation in the absence of testing, it is not apparent that the submission raises a question of law. In any event, the submission again fails to appreciate the distinction between the *outcome*, on the one hand, and yield values or data or results, on the other.

105 Insofar as the applicant submits that Dr Euston gave evidence that the overall outcome of the work could not have been accurately known in advance, it is not apparent that this raises a question of law. In any event, this evidence needs to be read in context. When so read, it is capable of referring to the *data* or *results* as distinct from the *outcome* of the activities. Dr Euston also gave evidence that A&B Mylec "would have anticipated a less than favourable outcome from their work with the FCCM". Dr Euston also stated that the "quality

of these coals is well documented" and he was "confident that A&B Mylec would have been aware of this work" (see [61] above).

106 Insofar as the applicant relies on the evidence of Mr Vorias to challenge the Tribunal's finding in the last sentence of [88], this does not appear to raise a question of law.

107 Insofar as the applicant submits that the Tribunal applied the wrong test by asking whether the outcome was *predictable*, we do not accept that submission. The Tribunal's statement in the last sentence of [88] that the experts agreed that the outcome was "predictable" merely reflected the evidence that the experts had given. The Tribunal went on to refer to the outcome being one that "could have been determined in advance", reflecting the statutory test. For the reasons given above, there was evidence to support that finding.

108 Insofar as the applicant submits that it was incorrect and irrelevant to determine whether the work *in fact* generated new knowledge, because the question is whether the experimental activities are conducted *for the purpose of* generating new knowledge, this takes the Tribunal's finding out of context. The Tribunal's finding at [88] reflected the evidence of the experts (discussed above). The Tribunal dealt elsewhere (at [90]) with the question whether the activities were conducted for the purpose of generating new knowledge.

109 For these reasons, the applicant's contentions in relation to the first issue are not made out.

The second issue

110 The second issue is whether the Tribunal erroneously substituted an alternative inquiry for that mandated by the statutory language in s 355-25(1)(a), which requires consideration of whether the outcome of activities "can only be determined by applying a systematic progression of work that: (i) is based on principles of established science; and (ii) proceeds from hypothesis to experiment, observation and evaluation, and leads to logical conclusions".

111 The applicant contends that the Reasons at [90] (see [85] above) reveal that the Tribunal did not apply the text of the statute, but determined the issue adversely to the applicant on the basis of an a priori assumption that whether the statutory criteria are satisfied depends upon the existence of "R&D plans or documentation". The Tribunal referred to two previous Tribunal decisions – *Rix's Creek Pty Ltd; Bloomfield Collieries Pty Ltd and Innovation Australia* [2017] AATA 645 (***Rix's Creek***) and *Docklands Science Park Pty Ltd and Innovation Australia* [2015] AATA 973 (***Docklands***) – as supporting that approach. The applicant submits that those decisions either should not have been regarded as stating any

principle of general application or were incorrectly decided insofar as they did seek to outline such a general principle.

112 The applicant submits that the correct approach is that identified by Tamberlin DP and Member Emert in *Re DBTL and Innovation Australia* (2013) 137 ALD 88 at [155]: "The purpose of a research and development activity may be established, for example, by oral testimony or contemporaneous documentation". The applicant also relies on *Commissioner of Taxation v Bogiatto* [2020] FCA 1139 at [100]-[101] per Thawley J for the proposition that the requirements for eligibility for the R&D tax offset can be established without the need for documentation.

113 In our view, the premise of the applicant's contention, namely that the Tribunal proceeded on an assumption that it was necessary for there to be R&D plans or documentation to satisfy the requirements of s 355-25(1), is not made out. The Tribunal did not state in [90] (or elsewhere) that the existence of R&D plans or documentation was essential to satisfy the requirements of s 355-25(1). In our view, when [90] of the Reasons is read in context, it is apparent that the reference to the absence of R&D plans or documentation merely reflected the facts of the present case rather than an assumption that such plans or documentation were required to satisfy the provision. The Tribunal's observation that there were no R&D plans or documentation was made in the course of considering the hypothesis put forward by Mr Vorias in his statement (see [31] above) and assessing whether the activities in question were activities whose outcome "can only be determined by applying a systematic progression of work that (i) is based on principles of established science; and (ii) proceeds from hypothesis to experiment, observation and evaluation, and leads to logical conclusions" (s 355-25(1)). In this context, the Tribunal made the observation that "there are no R&D plans or documentation to demonstrate the activities were carried out by applying a systematic progression of work based on principles of established science; or that they proceeded from the purported hypothesis to experiment, observation and evaluation, leading to logical conclusions". We take this to be a statement about the state of the evidence in the present case, rather than reflecting an assumption that such documentation is necessary to satisfy the provision. The absence of R&D plans or documentation was one factor taken into account by the Tribunal in reaching the conclusion that the relevant parts of s 355-25(1) were not satisfied.

114 Further, the Tribunal's citation of *Rix's Creek* and *Docklands* does not establish that it made the alleged error. Indeed, in *Rix's Creek* at [21], the Tribunal accepted that the creation and provision of documentation was *not* a statutory requirement. The point made by the Tribunal in that case was that "documentary evidence is an expected feature of an activity that is systematic, investigative and experimental" (at [21]).

115 For these reasons, we reject the applicant's contentions in relation to the second issue.

The third issue

116 The third issue is whether the Tribunal at [90] of the Reasons made findings that were not open on the evidence and applied a wrong legal standard, in considering whether the applicant's activities were "conducted for the purpose of generating new knowledge (including new knowledge in the form of new or improved materials, products, devices, processes or services)" within the meaning of s 355-25(1)(b).

117 The applicant's submissions focus on the following statement in [90] of the Reasons: "As Dr Euston characterised the activities, they merely built on the body of knowledge already available about the FCCM." The applicant submits that this finding was not open on the evidence. The applicant submits that, at no point in his written reports or oral evidence did Dr Euston say the proposition attributed to him or anything that would enable his evidence to be characterised as promulgating such an opinion. Further, the applicant submits that the Tribunal's reasoning suggests that it determined the issue by reference to that opinion (erroneously imputed to Dr Euston), or at the very least regarded it as a significant factor. The applicant submits that, even if such an opinion had been expressed by Dr Euston, the approach was incorrect, because the Tribunal failed to grapple with the proposition that "new knowledge" for the purposes of s 355-25(1)(b) includes improvements to existing processes. The applicant submits that, in determining the question of purpose adversely to the applicant, the Tribunal ignored substantial evidence that supported the applicant, including witness evidence that was unchallenged, and failed to give proper analysis as to the application of the actual text of the legislation to the evidence before it.

118 In our view, these submissions should be rejected. The submission that Dr Euston did not characterise the relevant activities as building on the body of knowledge already available about the FCCM, does not raise a question of law: see [94] above. In any event, the statement in the Reasons at [90] is consistent with Dr Euston's evidence: see the expert report of Dr Euston at [32], [38], [47] and [57]. For example, at [38] of his report, Dr Euston stated:

> In my opinion the activities as at that stage and described above (up to and including Activity iv)) were carried out to test the hypothesis, **to confirm previous knowledge of the FCCM** and to confirm the likely response of these banded deposits to conventional techniques of mining and coal beneficiation. The conclusion that I would have drawn at this stage is that the hypothesis is valid but not proven and that further work was justified.
>
> (Emphasis added.)

119 Insofar as the applicant submits that the Tribunal failed to grapple with the proposition that "new knowledge" for the purposes of s 355-25(1)(b) includes "new or improved … processes", this submission bears no relevant connection with the facts of the present case. On the findings of the Tribunal, the relevant activities were not undertaken for the purpose of generating new or improved processes; they were, rather, conducted for the purpose of reiterating the existing state of knowledge, by way of confirming the status of particular samples.

120 Insofar as the applicant submits that, in determining the question of purpose adversely to the applicant, the Tribunal ignored substantial evidence that supported the applicant, this does not raise a question of law: see [94] above.

121 In the applicant's amended outline of submissions it is submitted that, whether or not an activity in fact builds upon existing knowledge bears no necessary relationship to whether an activity is conducted for the *purpose* of generating new knowledge, which is the issue under s 355-25(1)(b). However, the nature of the activity conducted and the novelty and predictability of the results of that activity may elucidate whether a substantial purpose of an activity is the generation of new knowledge. It is artificial to discard the nature of an activity and what it achieves when assessing its purpose.

122 We note for completeness that in *Moreton* the Full Court found at [151] that s 355-25(1)(b) is "capable of applying, *depending on the circumstances*, to activities that are conducted for the purpose of generating new knowledge with respect to the application of an existing technology at a new site" (emphasis added). The gravamen of the Full Court's reasoning was that the generation of site specific knowledge by the application of the particular technology at issue in that case was capable of being "experimental activity" done for the "purpose of generating new knowledge" within the meaning of s 355-25(1). The facts of the present case are different from those in *Moreton* because the relevant activities did not involve applying existing technology to a new site for the purpose of generating new knowledge.

123 For these reasons, the applicant's contentions in relation to this issue are not made out.

The fourth issue

124 The fourth issue is whether the Tribunal misapplied s 355-25(1) by assessing the activities predominantly by reference to those undertaken in the 2012 year as opposed to taking into account the applicant's overall project, which would inevitably take several years, and consequently approached the application of s 355-25(2)(b) from an invalid starting point.

125 The applicant submits that this unduly narrow focus is evident from the Reasons at [13], [27], [28] and [90]. The applicant submits that: the evidence was that much of the 2012 year activities were directed to identifying locations for a bulk coal sampling program, to be undertaken by way of test pits known as costeans; those test pits were excavated (and filled in and rehabilitated) in late December 2012/January 2013, as the Tribunal noted at [92]; the purpose of the bulk coal sampling was that identified by Mr Vorias and set out by the Tribunal at [40] of the Reasons, namely "to undertake research and testing into the potential for new or improved mining methods and processing plant requirements"; on the evidence, that was done. The applicant submits that the Tribunal recognised, at [28], that activities in later years may shed light on the characteristics of the activities undertaken in the 2012 year, but then said at [74] that such evidence was given limited weight, despite suggesting at [90] that later years' activities "could be said to be in line with the purported hypothesis statement". The applicant submits that the meaning of that observation is unclear. The applicant submits that the evidence as to what was done in later years ought to have been regarded as being of considerable evidential significance as confirming the existence of a systematic progression of work undertaken for the purpose of generating new knowledge, as well as for other reasons.

126 The applicant submits that the Tribunal ought to have approached the inquiry necessitated by s 355-25(2)(b) by taking into account the purpose of the applicant's activities overall, of which the activities it had undertaken in the 2012 income year were but a part.

127 The applicant also challenges the Tribunal's conclusion at [91] that the activities fell within the exclusion in s 355-25(2)(b) on the basis that they were focussed on determining the size and quality of the deposit because they were "aimed at determining more precisely the expected variability in quality". The applicant submits that this appears to be a conclusion that the activities fell within s 355-25(2)(b)(iii). The applicant submits that it is very important to recognise that the exclusion imposes a purpose test; had the Tribunal properly applied the purpose test in s 355-25(1)(b), it *might* have concluded that the activities were

undertaken for the purpose of generating new knowledge in the form of new or improved processes for coal beneficiation; had it done so, it *might* have also decided that the activities were not undertaken for the prohibited purpose identified in s 355-25(2)(b)(iii). The applicant submits that, indeed, it is likely that a different conclusion would have been reached, but it is sufficient that a different conclusion *might* have been reached.

128 The applicant submits that there was considerable evidence upon which it might have been concluded that the applicant's activities were not undertaken for the purpose of determining the expected variability in quality of the deposit; as to the quality of the deposits, it was well known that the FCCM were high-ash, poor quality deposits; in addition, and contrary to what the Tribunal said at [91], there was no evidence capable of supporting the conclusion that the A&B Mylec and Sedgman work was undertaken for determining "the expected variability in quality" of the deposit; Dr Vince expressed this conclusion, at [20.1.2] and [20.1.3] of his report in relation to the survey activities and drilling but made no such suggestion in relation to the A&B Mylec and Sedgman work; Dr Euston characterised the A&B Mylec work (at [34] of his report) as following well-established experimental processes; he then said: "The aim of this testing is to reproduce, on a limited scale, the expected performance of the coal as it is transported by truck or conveyor between the mine and the CHPP [coal handling and processing plant]"; see also [37] of his report.

129 For the following reasons, we do not accept the applicant's submissions.

130 Insofar as the applicant submits that the Tribunal misapplied s 355-25(1) by assessing the relevant activities predominantly by reference to those undertaken in the 2012 year as opposed to taking into account the applicant's overall project, it is not established that the Tribunal misapplied the provision by limiting its consideration in this way. In support of this submission, the applicant relies on [13], [27], [28], [74] and [90] of the Reasons. We will refer to each of these paragraphs in turn. In [13], the Tribunal stated:

> It should be made clear that the applicant's vision of new ways to mine and process coal would only be realised, if at all, over an extended period. The applicant anticipated the project would span several income years. But it is also important to realise – for reasons we will explain – we are not evaluating the multi-year project as a whole. We must focus on whether the activities registered and conducted in the year of income in question qualify as "*core R&D activities*" or "*supporting R&D activities*" within the meaning of the IRD Act and the ITA Act. We explain that legislative regime next.

This paragraph evidences an awareness on the part of the Tribunal that the Project was a multi-year project. Insofar as the Tribunal clarified that it was not evaluating the multi-year project as a whole but rather was focussing on whether the registered activities met the definitions of "core R&D activities" or "supporting R&D activities", no error is shown in the Tribunal's approach: see s 27J of the IR&D Act.

131 In [27]-[28] of the Reasons, the Tribunal stated:

> 27. As dictated by s 27J of the IRD Act, consideration is to be had as to whether during the 2011-2012 year, all or part of the *registered activities* conducted were "*core R&D activities*" or "*supporting R&D activities*". Regard is not to be had to the *whole project* but rather to the *actual registered activities* conducted in the relevant registration year. This distinction was considered in *Industry Research and Development Board v Coal & Allied Operations Pty Ltd* [2000] FCA 979 at [46] to [49]. Lindgren J found that the Tribunal made an error when it referred to the project as a whole but had ultimately undertaken the correct analysis of the activities. At [47] Lindgren J said:
>
> > *Notwithstanding the AAT's dangerous frequent departures from the legislation's term ("activities") in favour of "project", I think it clear that the AAT applied the definition's criteria to the three categories of activity in dispute…*
>
> 28. We are not looking at the project as a whole when determining the issues in these proceedings but focus instead on the actual (core and supporting) activities undertaken in the 2011-2012 year. Evidence of activities which fall outside of the 2011-2012 year is only relevant in so far as it sheds appropriate light on the characteristics of the activities conducted in the 2011-2012 year.

These paragraphs do not indicate error in the Tribunal's approach; to the contrary, they indicate that the Tribunal correctly understood the enquiry required by s 27J of the IR&D Act. Moreover, the last sentence of [28] makes clear that the Tribunal did not exclude from consideration evidence about the activities conducted in later years.

132 In [74] of the Reasons, the Tribunal stated:

> In addition, the applicant tendered the witness statements of Dr Tobias Krull dated 20 October 2017 and Dr Peter J Bethell dated 23 October 2017. … No party required the witnesses for cross examination. We give these witness statements limited weight due to relevance. Both relate to activities conducted outside of the 2011-2012 year - namely, the work conducted by TUNRA in 2015 and the work conducted by Virginia Tech and Taggart.

Read in the context of [27]-[28], we see no error in the Tribunal's approach. Moreover, the weight to be given to particular evidence is a matter for the Tribunal and does not of itself give rise to a question of law.

133 In relation to [90] of the Reasons, this paragraph does not demonstrate that the Tribunal adopted an unduly narrow focus. For ease of reference we set out the last part of that paragraph again:

> It was only after the disappointing yet predicable results of A&B Mylec and Sedgman that the applicant turned its mind to new concepts to identify improvements to be made to existing coal mining and beneficiation processes, or new processes which could be developed to enable the mining, processing and handling (transporting) of the coal in the highly banded coal seams in the FCCM. It may be that those activities conducted by the applicant could be said to be in line with the purported hypothesis statement. However those activities fall outside of the 2011-2012 income year.

The Tribunal was here indicating that the applicant changed its conceptual approach after the disappointing (yet predictable) results of A&B Mylec and Sedgman, and that the activities conducted after that change were distinct from those undertaken during the 2012 year. There was evidence to support these findings, and the findings do not demonstrate any error in the Tribunal's approach.

134 Accordingly, we do not accept the submission that the Tribunal misapplied s 355-25(1) by assessing the relevant activities predominantly by reference to those undertaken in the 2012 year as opposed to taking into account the applicant's overall project. It follows that we do not accept that the Tribunal consequently approached the application of s 355-25(2)(b) from an invalid starting point.

135 Some of the applicant's submissions appear to proceed on the basis that if an activity was undertaken for the purpose of generating new knowledge, it cannot have been exploration or drilling for the purposes of determining more precisely the location, size or quality of deposits. However, there is no mutual exclusivity between sub-s (1) and sub-s (2) of s 355-25. An activity might be conducted for the purpose of generating new knowledge, but if the activity in question is prospecting, exploring or drilling for minerals and the new knowledge in question concerns the location, size or quality of deposits the exclusion in s 355-25(2)(b) may apply.

136 To the extent that the applicant's submissions point to evidence in support of its contention that the activities did not fall within the exclusion in s 355-25(2)(b), the submissions do not raise a question of law. As stated above, the weight to be accorded to the evidence adduced by the parties is a matter for the Tribunal and does not of itself give rise to a question of law. The Tribunal's finding in relation to s 355-25(2)(b) (at [91] of the Reasons) was supported by

the Tribunal's earlier factual findings in relation to the relevant activities at [85]-[88]. Those findings were consistent with the evidence of Dr Euston, who explained why, in his view, the activities were intended to produce information about the size, variability and quality of the coal deposits at the FCCM: see [61] above.

137 For these reasons, we reject the applicant's contentions in relation to this issue.

The fifth issue

138 The fifth issue is whether the Tribunal erred by ignoring relevant evidence, and whether it failed to give adequate reasons to the extent that it did not accept the applicant's evidence, and whether it failed to make findings on material questions of fact.

139 As to the Tribunal ignoring relevant evidence, the applicant submits that the evidence of Mr Vorias appears to have been accepted (there is no statement in the Reasons that it was not). The applicant submits that that evidence (extracted in part at [39] and [40] of the Reasons) was to the effect that:

(a) Mr Vorias and the board of the applicant recognised in about July 2011 that it might be possible "for improvements to be made to existing coal mining and beneficiation processes, or new processes could be developed, that would enable the mining, processing and handling (transporting) of highly banded coal seams known to exist in the Bowen Basin … by achieving a combination of (a) the extraction of material with a higher yield of coal to waste than had been done previously, (b) the development of a crushing and washing plant configuration that would operate differently from those that had previously been used and that (c) would result in a product able to be handled and transported, which had not previously been thought possible."

(b) Mr Vorias identified that there was a need for a bulk coal sampling program and the excavation of test pits to obtain samples which would enable experiments to verify or disprove the hypothesis (which was, in summary, that improvements could be made to existing processes or new processes developed of the kind set out above).

(c) The main purpose of the surveys and drilling work undertaken in the 2012 year was to help in identifying the "best locations within the EPC area at which the test pits could be excavated".

140 The applicant submits that Mr O'Malley, who was not cross-examined, and whose evidence was admitted without objection, gave evidence that:

(a) The applicant's activities were carried out to determine whether new mining and beneficiation processes could be developed to enable coal within the FCCM to be mined, processed and handled economically.

(b) The survey work and drilling results enabled identification of the best sites to undertake test excavations, one of the main purposes of them being to identify areas for test excavations and that "[t]his had to be done to obtain bulk samples of the coal for testing and experiments directed to determining whether new mining and processing methods could be developed …".

141 As to the Tribunal failing to give adequate reasons, the applicant submits that, to the extent that the Tribunal may have not accepted the evidence of Mr Vorias, or the unchallenged evidence of Mr O'Malley, it failed to give adequate reasons for doing so.

142 As to the Tribunal failing to make findings on material questions of fact, the applicant submits that:

(a) The Tribunal failed to determine whether or not it could have been known or determined in advance, that improvements could (or could not) be made to existing mining and beneficiation methods, or new processes developed (or not), to enable the mining, processing and handling of highly banded coal seams known to exist in the Bowen Basin and elsewhere to be achieved economically.

(b) The Tribunal failed to determine whether or not new or improved mining and beneficiation methods or processes, that would enable the mining, processing and handling of highly banded coal seams known to exist in the Bowen Basin and elsewhere to be achieved economically, could only be identified by applying a systematic progression of work: (i) that is based on principles of established science; and (ii) proceeds from hypothesis to experiment, observation and evaluation, and leads to logical conclusions.

143 For the following reasons, we do not accept these submissions.

144 Contrary to the applicant's submissions, the Tribunal did not ignore the evidence of Mr Vorias and Mr O'Malley, but referred to it as relevant to the issues as they arose for consideration: see the Reasons at [40], [43], [73], [86]-[87] and [90]. The Tribunal was not under a duty to refer to every piece of evidence and provide reasons in relation to every piece

of evidence. It had to consider evidence critical to the issues in dispute: see generally *Keith v Gal* [2013] NSWCA 339 at [109]-[119] per Gleeson JA. The Tribunal carried out this duty.

145 Insofar as the applicant submits that the Tribunal failed to make findings on certain material questions of fact, the matters raised by the applicant did not arise for consideration on the Tribunal's findings. They involve abstracted propositions that do not align with the decisive issues that arose for determination by the Tribunal in seeking to characterise each of the relevant activities by reference to the statutory criteria in s 355-25.

146 We therefore reject the applicant's contentions in relation to this issue.

Question of law 12 and the associated grounds

147 By question of law 12 and grounds 8.1 and 8.2, the applicant contends, in summary, that the Tribunal erred by failing to consider a submission advanced by the applicant below to the effect that activities undertaken in the 2012 year should be regarded as "supporting R&D activities" within the meaning of s 355-30 if they fell within s 355-25(2)(b), on the basis that they were directly related to and carried out for the dominant purpose of supporting core R&D activities in later years, namely the work by Taggart and Virginia Tech. The applicant also contends that the Tribunal erred (at [92] of the Reasons) in characterising the applicant's argument as being that drilling and survey activities were supporting R&D activities only on the basis that they were undertaken for the dominant purpose of supporting a core R&D activity that was limited to excavation of the costeans.

148 The applicant submits that the Tribunal should have dealt with the following submission put to it in the applicant's outline of closing submissions:

> 107. Further, or in the alternative, if the drilling and surveys were undertaken for an excluded purpose, they were undertaken for the dominant purpose of supporting core activities and in particular the work of A&B Mylec and Sedgman, and/or other testing activities. The A&B Mylec and Sedgman work, and the Taggart work, could not be described as "prospecting, exploring or drilling for minerals" for any of the excluded purposes.
>
> 108. The Applicant's case was opened on this basis and although the drilling is not identified as a supporting activity in the registration application, it is submitted that the applicant's position is clear. Supporting activities need not occur in the same year as the core activities that they support: see s 27J(1)(c)(iii) of the IR&D Act. The absence of an overseas finding for the Taggart work does not exclude that work from being a core activity within the statutory definition.

(Footnote omitted.)

149 The Board accepts that an activity may be found to be a "supporting R&D activity" even though it was registered as a "core R&D activity" (T52). Further, it is clear from the text of s 27J(1)(c)(iii) of the IR&D Act that an activity may be found to be a "supporting R&D activity" with respect to a "core R&D activity" to be conducted in a *later year*.

150 For the following reasons, we do not accept the applicant's contentions.

151 First, it is apparent from the Tribunal's findings that the work conducted by Taggart and Virginia Tech represented a conceptually different approach, devised only after the disappointing (yet predictable) results from the A&B Mylec and Sedgman work: see [89] of the Reasons. The proposition that the activities conducted during the 2012 year were "directly related" to the Taggart and Virginia Tech work, and were for the "dominant purpose" of supporting that work, is not supported by, and indeed is contrary to, the Tribunal's findings.

152 Secondly, the way the Tribunal dealt with the issue of "supporting R&D activities" (at [92]) reflected the way the argument was presented to the Tribunal. In the first sentence of [92], the Tribunal stated: "The applicant contends in the alternative that the drilling and survey activities were 'supporting R&D activities' because they were conducted for the dominant purpose of supporting a core R&D activity, namely, the excavation of costeans." This reflected the applicant's outline of closing submissions at paragraphs 112-114, comprising the applicant's conclusions in relation to "supporting R&D activities". Those paragraphs were as follows:

> **Conclusions – Supporting R&D Activities**
>
> 112. At Activity 1.1.1 of its Registration Application, the Applicant described supporting R&D Activities, including literature search and review of existing mining and beneficiation processes and technologies, the SkyTEM and 2D surveys.
>
> 113. If the Applicant's drilling or surveying activities are determined by this Tribunal not to be core R&D activities, either because the Tribunal determines that they do not fall within the definition in s.355-25(1) or because they are determined by the Tribunal to be excluded activities under s.355-25(2), then the Applicant submits that these activities fit comfortably within the definition of 'supporting R&D activities' in s.355-30, as they directly related to core R&D activities and were undertaken for the dominant purpose of supporting those core R&D activities.
>
> 114. **By way of example, all drilling work that is the subject of this application was performed for the purpose of identifying optimal locations for the costeans** (which in turn were excavated in order that samples could be sent for testing to determine whether a new kind of coal

> handling and processing plant could be developed) and to produce core samples to be similarly tested, and to validate the SkyTEM results.
>
> (Footnote omitted; emphasis added.)

153 The applicant's example in paragraph 114 of its outline reflected the evidence of Mr Vorias that the "main purpose" of the surveys and drilling was "to help in identifying the best locations within the EPC area at which the test pits could be excavated" (see [32] above). While it is true that paragraph 114 of the applicant's outline of closing submissions was provided "[b]y way of example", and therefore was not exhaustive, it is unsurprising in the circumstances (given the evidence of Mr Vorias and the Tribunal's other findings) that the Tribunal focussed on this example in its consideration of "supporting R&D activities" in [92] of the Reasons.

154 For these reasons, the applicant's contentions in relation to question of law 12 and the associated grounds are not made out.

Conclusion

155 It follows from the above that the appeal is to be dismissed. It was common ground at the hearing that costs should follow the event. Accordingly, we will also order that the applicant pay the Board's costs of the appeal, as agreed or taxed.

I certify that the preceding one hundred and fifty-five (155) numbered paragraphs are a true copy of the Reasons for Judgment of the Honourable Justices Logan, Griffiths and Moshinsky.

Associate:

Dated: 23 April 2021